Domain & Range

What can go in? What can come out?

Answer Key with complete solutions

The big idea

A function is a machine: you put in an x, it gives back one y.

- The **domain** is every x you are **allowed to put in**. (inputs, the horizontal direction)
- The **range** is every y that can **actually come out**. (outputs, the vertical direction)

Simple case first. For $f(x) = \sqrt{x}$, try a few inputs: $f(9) = 3$, $f(0) = 0$, but $f(-4)$ is not a real number. So the domain is $x \geq 0$. And since a square root is never negative, the outputs are $y \geq 0$. Testing a few easy numbers is always a good way to start.

1 Three ways to write a set of numbers

In words	Inequality	Interval notation	Set-builder
from -2 to 5, including -2, not 5	$-2 \leq x < 5$	$[-2, 5)$	$\{x \mid -2 \leq x < 5\}$
greater than 3	$x > 3$	$(3, \infty)$	$\{x \mid x > 3\}$
at most -1	$x \leq -1$	$(-\infty, -1]$	$\{x \mid x \leq -1\}$
all real numbers	—	$(-\infty, \infty)$	$\mathbb{R}$
every number except 2	$x \neq 2$	$(-\infty, 2) \cup (2, \infty)$	$\{x \mid x \neq 2\}$

Square bracket [ ]: the endpoint is **included** (closed dot •, $\leq$ or $\geq$).

Parenthesis ( ): the endpoint is **not included** (open dot ∘, $<$ or $>$).

∞ is not a number you can reach, so it **always** gets a parenthesis.

$\cup$ means "or" — it joins two pieces.

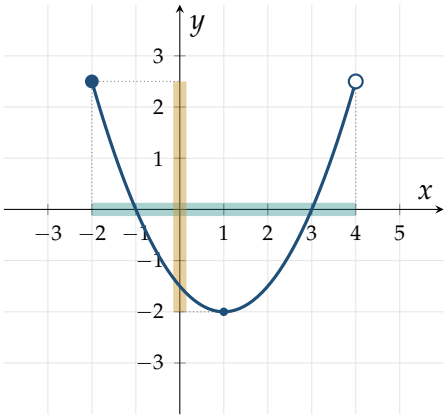
2 From points, a table, or a mapping

Example 1

$\{(-1, 4),\ (0, 2),\ (3, 4),\ (5, -1)\}$ Domain: list the x's: $\{-1, 0, 3, 5\}$. Range: list the y's: $\{-1, 2, 4\}$.

List each value **once**, from smallest to largest (4 appears twice but is written once).

3 From a graph: look at the shadows



Shine a light from above: the graph's **shadow on the x-axis** (teal) is the **domain**. Shine it from the side: the **shadow on the y-axis** (gold) is the **range**.

Read the domain left to right: from $x = -2$ (closed dot, included) to $x = 4$ (open dot, not included):

$$\text{Domain} = [-2, 4).$$

Read the range bottom to top: the lowest point is the vertex, $y = -2$. The highest y is 2.5 — the open dot does *not* reach it, but the closed dot at $(-2, 2.5)$ does:

$$\text{Range} = [-2, 2.5].$$

Watch out: an **arrow** means the graph keeps going, so that side goes to ∞ or $-\infty$. An open dot only leaves out that *one point* — check whether another part of the graph reaches the same y.

4 From an equation: what could go wrong?

Domain: start with all real numbers, then remove only what breaks

1. Division by zero	a denominator can't be 0	$\dfrac{1}{x-3}$: $x \neq 3$
2. Even roots of negatives	what is under $\sqrt{\ }$ must be ≥ 0	$\sqrt{x+5}$: $x \geq -5$
3. Real-world limits	time, length, people can't be negative	$t \geq 0$, n a whole number

Nothing else breaks: polynomials, absolute value and **odd** roots like $\sqrt[3]{x}$ have domain $\mathbb{R}$.

Range: where can the outputs go? Use the shape of the graph.

Function	Look at…	Range
$a(x-h)^2 + k$	the vertex (h, k): up if $a > 0$, down if $a < 0$	$[k, \infty)$ or $(-\infty, k]$
$a\sqrt{x-h} + k$	the starting point (h, k)	$[k, \infty)$ if $a > 0$, $(-\infty, k]$ if $a < 0$
$a\lvert x-h\rvert + k$	the vertex (h, k)	$[k, \infty)$ or $(-\infty, k]$
$\dfrac{a}{x-h} + k$	the horizontal asymptote $y = k$	every y except k
$mx + b$, $m \neq 0$	—	$\mathbb{R}$

Example 2 · a quadratic in standard form

$f(x) = -x^2 + 6x - 5$. Domain: a polynomial, so $\mathbb{R}$.
Vertex: $x = -\frac{b}{2a} = -\frac{6}{-2} = 3$, and $f(3) = -9 + 18 - 5 = 4$. It opens down ($a < 0$), so the highest output is 4: Range $= (-\infty, 4]$.

Example 3 · a square root

$g(x) = \sqrt{2x-6}+1$. Domain: $2x-6 \geq 0 \Rightarrow x \geq 3$, so $[3,\infty)$.
Range: the square root starts at 0 and grows, so g starts at $0+1=1$ and grows: $[1,\infty)$. Simple case check: $g(3)=1$, $g(5)=3$. ✓

Example 4 · a rational function — solve for x to find the range

$h(x) = \dfrac{2x+1}{x-4}$. Domain: $x \neq 4$.
Range: set $y = \dfrac{2x+1}{x-4}$ and solve for x: $yx-4y = 2x+1 \Rightarrow x(y-2) = 4y+1 \Rightarrow x = \dfrac{4y+1}{y-2}$.
This works for every y **except** $y=2$. So Range $= \{y \mid y \neq 2\}$ — the horizontal asymptote.

Common mistakes: (1) Writing the domain of $\sqrt{x-4}$ as $x > 4$ — it is $x \geq 4$, because $\sqrt{0}=0$ is fine. (2) Saying the domain of $\frac{1}{x^2+1}$ excludes something — x^2+1 is never 0, so the domain is $\mathbb{R}$. (3) Mixing up domain (x) and range (y).

5 Practice

A. Sets, tables and notation EASY

1. Find the domain and range: $\{(-2,3),\ (0,5),\ (1,3),\ (4,-1)\}$

> Domain $\{-2,0,1,4\}$; Range $\{-1,3,5\}$ (write 3 only once).

2. Find the domain and range from the table. Is it a function?

x	-3	-1	0	2	5
y	7	7	1	-2	0

> Domain $\{-3,-1,0,2,5\}$; Range $\{-2,0,1,7\}$. Yes, a function: no x repeats (a repeated y is fine).

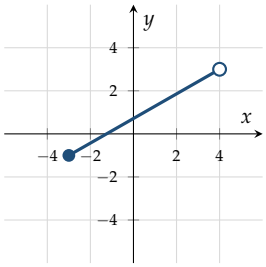
3. Write in interval notation: (a) $-4 \le x < 1$ (b) $x > -2$ (c) $x \le 0$ or $x > 3$ (d) all reals except -5

> (a) $[-4,1)$ (b) $(-2,\infty)$ (c) $(-\infty,0]\cup(3,\infty)$ (d) $(-\infty,-5)\cup(-5,\infty)$

B. From a graph MEDIUM

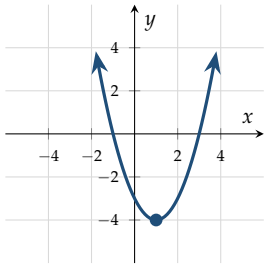
Give the domain and range of each graph. Arrows mean the graph continues.

4.



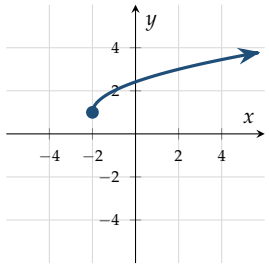
$D = [-3,4),\ R = [-1,3)$

5.



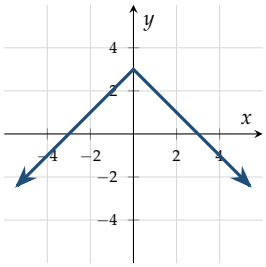
$D = \mathbb{R},\ R = [-4,\infty)$

6.



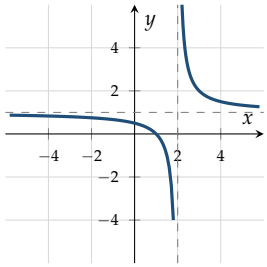
$D = [-2,\infty),\ R = [1,\infty)$

7.



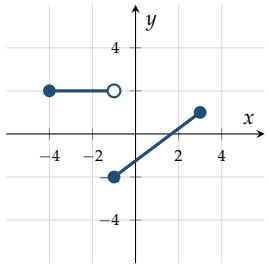
$D = \mathbb{R},\ R = (-\infty,3]$

8.



$D : x \neq 2,\ R : y \neq 1$ (asymptotes)

9.



$D = [-4,3],\ R = [-2,1]\cup\{2\}$

Note on 9: the left piece is flat at $y = 2$, so 2 is in the range even though the open dot leaves out $(-1,2)$; the right piece covers y from -2 to 1.

C. Domain from an equation MEDIUM

10. $f(x) = 3x^2 - 5x + 1$

$\mathbb{R}$ — a polynomial.

11. $g(x) = \sqrt{x-6}$

$x - 6 \geq 0$: $[6, \infty)$

12. $h(x) = \sqrt{10 - 2x}$

$10 - 2x \geq 0 \Rightarrow x \leq 5$: $(-\infty, 5]$

13. $k(x) = \dfrac{x+1}{x-7}$

$x \neq 7$: $(-\infty, 7) \cup (7, \infty)$

14. $m(x) = \dfrac{5}{x^2 - 9}$

$x^2 \neq 9$, so $x \neq \pm 3$: $(-\infty, -3) \cup (-3, 3) \cup (3, \infty)$

15. $p(x) = \sqrt[3]{x-4}$

$\mathbb{R}$ — odd roots accept negatives.

16. $q(x) = \dfrac{\sqrt{x+2}}{x-3}$

Need $x \geq -2$ **and** $x \neq 3$: $[-2, 3) \cup (3, \infty)$

17. $r(x) = \dfrac{4}{x^2 + 1}$

$x^2 + 1 \geq 1$, never 0: $\mathbb{R}$

D. Domain and range from an equation MEDIUM HARD

18. MEDIUM $f(x) = (x-3)^2 + 2$

Domain $\mathbb{R}$. Vertex $(3, 2)$, opens up: Range $[2, \infty)$.

19. MEDIUM $f(x) = -2x^2 + 8x - 1$

Domain $\mathbb{R}$. Vertex at $x = -\frac{8}{2(-2)} = 2$, $f(2) = -8 + 16 - 1 = 7$. Opens down: Range $(-\infty, 7]$.

20. MEDIUM $f(x) = \sqrt{x+4} - 3$

Domain: $x + 4 \geq 0$, so $[-4, \infty)$. Starts at $(-4, -3)$ and rises: Range $[-3, \infty)$.

21. HARD $f(x) = -\sqrt{9 - x} + 5$

Domain: $9 - x \geq 0 \Rightarrow x \leq 9$: $(-\infty, 9]$. The root is ≥ 0, so $-\sqrt{} \leq 0$ and $f(x) \leq 5$: Range $(-\infty, 5]$. Check: $f(9) = 5$, $f(0) = -3 + 5 = 2$.

22. MEDIUM $f(x) = |x+1| - 4$

Domain $\mathbb{R}$. Vertex $(-1,-4)$, V opens up: Range $[-4,\infty)$.

23. MEDIUM $f(x) = \dfrac{2}{x+3} - 1$

Domain $x \neq -3$. Horizontal asymptote $y = -1$ (the fraction $\frac{2}{x+3}$ is never 0): Range $y \neq -1$, i.e. $(-\infty,-1) \cup (-1,\infty)$.

24. HARD $f(x) = \dfrac{3x-1}{x+2}$ (find the range by solving for x)

Domain $x \neq -2$. $y(x+2) = 3x-1 \Rightarrow yx - 3x = -1-2y \Rightarrow x(y-3) = -1-2y \Rightarrow x = \dfrac{-1-2y}{y-3}$.
Every y works except $y = 3$: Range $(-\infty,3) \cup (3,\infty)$.

E. Real-world domain and range MEDIUM

25. Movie tickets cost \$12 each. A group of at most 8 friends buys n tickets, and the cost is $C(n) = 12n$. Give a reasonable domain and range.

n must be a whole number: Domain $\{0,1,2,\ldots,8\}$. Range $\{0,12,24,\ldots,96\}$.
(A list, not an interval — you can't buy 2.5 tickets.)

26. A ball is thrown upward. Its height in feet after t seconds is $h(t) = -16t^2 + 64t$, until it hits the ground. Find the domain and range.

It hits the ground when $h(t) = 0$: $-16t(t-4) = 0$, so $t = 0$ or $t = 4$. Domain $[0,4]$.
Highest point at $t = -\frac{64}{2(-16)} = 2$: $h(2) = -64 + 128 = 64$. Range $[0,64]$.

F. Challenge HARD

27. $f(x) = \sqrt{x^2 - 9}$

Need $x^2 - 9 \geq 0$, i.e. $x^2 \geq 9$: $x \leq -3$ or $x \geq 3$. Domain $(-\infty,-3] \cup [3,\infty)$.
$x^2 - 9$ takes every value ≥ 0, so Range $[0,\infty)$.

28. $f(x) = \dfrac{x^2 - 4}{x - 2}$

Domain $x \neq 2$. For $x \neq 2$, $f(x) = \dfrac{(x-2)(x+2)}{x-2} = x + 2$ — a line with a **hole** at $(2, 4)$.
So every y except 4: Range $(-\infty, 4) \cup (4, \infty)$.

29. $f(x) = \sqrt{16 - x^2}$ (Hint: what shape is the graph?)

Need $16 - x^2 \geq 0$: Domain $[-4, 4]$. The graph is the top half of the circle $x^2 + y^2 = 16$, so it goes from 0 up to $\sqrt{16} = 4$: Range $[0, 4]$.